AES17 Scaling and Farina Method Calibration

Part 1: The AES17 Farina Scaling of 3 dB

The **AES17 standard** and the **Farina log-swept sine method** apply a **+3 dB scaling factor** to resolve a core conflict in audio metering: the mathematical definition of average power (RMS) versus human-centric visual readability.

By default, the RMS (Root Mean Square) value of a pure sine wave is mathematically **3.01 dB lower** than its peak value. The AES17 convention shifts the RMS scale upward by exactly +3 dB so that **a full-scale sine wave reads 0 dB on both peak and RMS meters**.

The Underlying Mathematical Problem

For any standard sine wave, the peak-to-RMS relationship is derived from the square root of 2:

$$RMS = Peak / \sqrt{2} \approx 0.707 \times Peak$$

When converted into the logarithmic decibel full-scale digital domain (dBFS):

$$20 \log_{10}(0.707) \approx -3.01 \, dB$$

Without any scaling adjustments, if you generate a maximum-amplitude digital sine wave that touches the absolute ceiling of your digital system (0 dBFS Peak), a traditional RMS meter will read **-3.01 dBFS RMS**.

How AES17 and Farina Resolve This

To make equipment calibration, software metering, and acoustic transfer analysis more intuitive, the **AES17-1998** standard explicitly defines full scale (1 FS) based on the RMS amplitude of a sine wave rather than its absolute peak value.

- **The Alignment (Parity):** By artificially injecting a +3.01 dB gain offset into the RMS calculation path, **a pure sine wave's peak value and its RMS value become identical on the meter**.
- **Angelo Farina's Contribution:** In acoustic and speaker testing, Angelo Farina popularized the *logarithmic swept-sine technique* to measure impulse responses and total harmonic distortion (THD). Because Farina's method sweeps a continuous sine wave through the frequency spectrum to isolate linear system responses from non-linear distortions, it relies directly on the AES17 definition. Applying the +3 dB scaling factor ensures that the input power excitation remains perfectly scaled and aligned with digital clipping thresholds (0 dBFS).

Direct Metering Comparison

Signal Type (at 0 dBFS Peak)	Standard RMS Metering	AES17 / Farina Metering (+3 dB Shift)
Pure Sine Wave	-3.01 dBFS	**0.00 dBFS** (Perfect Parity)
Square Wave	0.00 dBFS	**+3.01 dBFS** (Signals clipping/ exceeding sine power)
Pink Noise	≈ -12.00 dBFS (variable)	≈ -9.00 dBFS

Why This Matters in Practice

- **Prevents Calibration Confusion:** Audio engineers using test signals (which are almost exclusively sine waves) can match levels without remembering to subtract 3 dB when switching eyes between a Peak meter and an RMS meter.

- **Exposes Square-Wave Clipping:** Because a square wave has a crest factor of 1 (0 dB difference between peak and RMS), it contains more raw power than a sine wave. Under the AES17/Farina framework, a heavily clipped square-wave signal will actively meter above 0 dB (pushing toward +3 dBFS), immediately flagging a dangerous over-saturation of power that standard full-scale calculations mask.

Part 2: Frequency Response Derivation — H1 Estimator vs. Farina Method

The **H1 estimator yields the correct physical SPL level**, while the uncompensated Farina method produces an artificially inflated result that is exactly 3.01 dB too high.

To bring both measurements into agreement, you must **subtract 3.01 dB (or divide the linear magnitude by √2) from your uncompensated Farina method results**.

Why the H1 Estimator is Correct

The H_1 estimator calculates the frequency response function (FRF) by dividing the cross-power spectral density (S_{xy}) by the auto-power spectral density of the input (S_{xx}):

$$H_1(f) = S_{xy}(f) / S_{xx}(f)$$

Because it maps the exact physical ratio of input power to output power, it maintains strict mathematical scaling. An impulse response derived from a properly calibrated H_1 estimator directly reflects physical quantities (e.g., pressure output per Volt input, or Pascal per Volt).

Why the Uncompensated Farina Method is 3 dB Higher

The uncompensated Farina method leverages a specific implementation shortcut: it deconvolves the recorded exponential sine sweep by convolving it with a time-reversed, amplitude-equalized version of the original excitation chirp.

The +3.01 dB shift in the uncompensated Farina method occurs due to **RMS-to-Peak Scaling Parity**:

- To make digital metering easier, the uncompensated Farina inverse filter uses the **AES17 definition of full-scale power**.

- Under AES17, a digital sine wave with a peak value of 0 dBFS is scaled up by +3.01 dB so that its Root Mean Square (RMS) value reads as 0 dB instead of -3.01 dB.
- When the deconvolution convolution executes without dropping this metering offset, that energy boost is carried directly into the resulting impulse response. When converted to the frequency domain, it registers as an extra +3 dB of SPL.

How to Apply the Correction Factor

To match the uncompensated Farina calculations to the absolute physical levels of your H_1 estimator, apply one of the following scaling modifications:

- **In the Logarithmic Domain (Decibels):**
  `SPL_Corrected = SPL_Farina - 20*log10(√2) ≈ SPL_Farina - 3.01 dB`
- **In the Linear Domain (Magnitude Spectral Data):**
  `H_Corrected(f) = H_Farina(f) / √2 ≈ H_Farina(f) × 0.7071`
- **In the Inverse Filter (Prior to Deconvolution):**
  If you have access to the raw script generating the Farina inverse filter, scale the entire time-domain array of the inverse filter by multiplying it by 0.7071 before convolving it with your recorded room sweep. This natively removes the scaling offset at the root level.